

THE UNIVERSITY OF DODOMA



COLLEGE OF EARTH SCIENCE AND ENGINEERING

DEPARTMENT OF PETROLEUM AND ENERGY

FINAL YEAR PROJECT REPORT PROPOSAL

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PROJECT TITLE: DESIGN AND DEVELOPMENT OF A SMART DOCUMENTATION AND REPORTING WEB BASED SYSTEM FOR PETROLEUM SHORE TANK INSPECTION OPERATIONS AT PETROLEUM BULKY PROCUREMENT AGENCY (PBPA)

CERTIFICATION

The undersigned certify that they have read and hereby recommend for examination by the university of Dodoma a final year project entitled: DESIGN AND DEVELOPMENT OF A SMART DOCUMENTATION AND REPORTING WEB BASED SYSTEM FOR PETROLEUM SHORE TANK INSPECTION OPERATIONS AT PETROLEUM BULKY PROCUREMENT AGENCY (PBPA) in partial fulfilment of the requirements for the degree of bachelor of science (Petroleum Engineering) of the University of Dodoma

Mr. Goodluck Oyolla

(Supervisor)

Date

DECLARATION

I Erick Vedasto Muhanuzi, declare that this final year project is my own original work and that it has not been presented to any other university for a similar or any other degree award.

Signature

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to Almighty God for granting me the strength, knowledge, and perseverance to complete this project successfully.

I extend my deepest appreciation to my academic supervisor for their guidance, constructive feedback, and continuous support throughout the project period. Their expertise and encouragement were invaluable in shaping this work.

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Finally, I acknowledge everyone who contributed directly or indirectly to the success of this project

ABSTRACT

The petroleum industry requires accurate measurement, documentation, and reporting of petroleum quantities to ensure operational efficiency, accountability, and regulatory compliance. At the Petroleum Bulk Procurement Agency (PBPA), petroleum shore tank inspection activities are primarily conducted using manual documentation methods and basic digital tools such as spreadsheets, which are time-consuming, prone to human error, and inefficient for real-time reporting and data management. This study aimed to design and develop a smart web-based documentation and reporting system to support petroleum shore tank inspection activities conducted by PBPA inspectors.

The study employed the Agile software development methodology to allow iterative system development, user feedback integration, and continuous system improvement. Data collection methods included interviews, observation, document review, and process mapping to analyse the existing manual processes and identify system requirements. The proposed system was developed using HTML, CSS, JavaScript, PHP, Python, and MySQL database technologies, following a three-tier architecture design consisting of presentation, application, and data layers.

The developed system integrates automated petroleum quantity calculations, secure data storage, user authentication, and automated report generation modules to enhance inspection efficiency and accuracy. System testing involved unit testing, integration testing, and user acceptance testing (UAT) with PBPA inspectors to validate functionality and usability. Results indicate that the system improves reporting accuracy, reduces preparation time, enhances data accessibility, and supports better decision-making compared to manual methods.

The study contributes to PBPA's digital transformation initiatives by providing a scalable and secure solution for petroleum inspection documentation and reporting in Tanzania

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LIST OF ABBREVIATIONS

API: Application Programming Interface

CSS: Cascading Style Sheets

ERD: Entity Relationship Diagram

HTML: HyperText Markup Language

IQ: Information Quality

MIS: Management Information Systems

MYSQL: Structured Query Language

PBPA: Petroleum Bulk Procurement Agency

PHP: Hypertext Preprocessor

STS: Socio-Technical Systems

TAM: Technology Acceptance Model

UAT: User Acceptance Testing

UI: User Interface

VSCode: Visual Studio Code

CHAPTER ONE

1. INTRODUCTION

The petroleum industry relies heavily on accurate measurement, documentation, and reporting of petroleum quantities to ensure accountability, operational efficiency, and regulatory compliance. Shore tank inspections play a critical role in verifying petroleum volumes received and stored at oil depots. However, manual documentation methods often lead to inefficiencies, delays, and errors that may affect decision-making processes.

With the advancement of information technology, web-based systems have become essential tools for improving data management, reporting accuracy, and accessibility across multiple operational locations. Therefore, this project focuses on developing a smart web-based documentation and reporting system to enhance petroleum shore tank inspection processes conducted by PBPA inspectors.

1.1 Background of the study

The modern petroleum industry generates large volumes of inspection and operational data that require accurate documentation, analysis, and reporting. Shore tank inspections are essential for ensuring safety, accurate volume determination, product accountability, and regulatory compliance. However, at the Petroleum Bulk Procurement Agency (PBPA), inspection data is still handled using manual methods and basic digital tools, making the reporting process time-consuming, inefficient, and prone to human error.

PBPA inspectors conduct initial, provisional, and final shore tank measurements at oil marketing company depots to verify received volumes and detect losses or gains. Inefficient reporting practices delay decision-making and negatively affect operational and managerial efficiency, inventory control, and supply planning. Studies show that digital database systems and web-based reporting platforms improve data accuracy, reduce errors, and support efficient handling of large datasets (Appigatla, MySQL 8 Cookbook. Packt Publishing., 2018); (Sotnik, 2023); (Michail, 2022.)

Web-based management information systems provide structured reporting frameworks, real-time access to data, and improved collaboration among inspectors across different locations (Deo, 2023); (Laudon, 2016). Technologies such as PHP/HTML, Python, and MySQL enable secure, scalable, and user-friendly reporting solutions while supporting regulatory compliance through access control and audit trails (Tatroe, 2020.)

Therefore, the development of a smart documentation and reporting web based system for petroleum shore tank inspections at PBPA is necessary to enhance efficiency, accuracy, transparency, and decision-making, while supporting PBPA's mandate to ensure a reliable and accountable petroleum supply in Tanzania.

1.2 Problem statement

Petroleum shore tank inspection is one of the activities (operations) conducted by Petroleum Bulk Procurement Agency (PBPA). These inspection operations involve recording, documenting and reporting petroleum shore tank measured data, that are currently documented and reported using manual methods and basic tools such as spreadsheets and reporting through WhatsApp platform. This approach is time-consuming, prone to human errors, and makes data consolidation and standardization difficult across different ports (Deo, 2023). Consequently, inspection reports may be delayed, inconsistent, and less reliable for timely operational and manageable decision-making (Laudon, 2016). Therefore, there is a need for a smart documentation and reporting web based system to improve the accuracy, efficiency, and real-time reporting to PBPA.

1.3 Project objectives

1.3.1 Main objective

To design and develop a smart documentation and reporting web based system to support Petroleum shore tank inspection operations conducted by Petroleum Bulky Procurement Agency (PBPA).

1.3.2 Specific objectives

- i. To analyse the existing manual processes for petroleum shore tank inspection operations at PBPA.
- ii. To develop a smart documentation and reporting web-based system for petroleum shore tank inspection operations at PBPA.
- iii. To test an automated reporting module.

1.4 Scope of the study

The study focuses on the design and development of a smart documentation and reporting web based system for petroleum shore tank inspection operations by PBPA. The system is designed for use by PBPA inspectors during inspection operations.

The study covers digital documentation and reporting of inspection data. This system will also involve integration of security protocols to store, protect and manage inspection documents with controlled access.

The scope further includes system design, development, and testing to ensure accurate data storage and reporting.

However, the study does not cover physical tank gauging techniques, laboratory testing of petroleum samples, automated sensor integration or operational systems used internally by oil marketing companies.

1.5 Significance of the study

This study is significant to the Petroleum Bulk Procurement Agency (PBPA) as it provides a smart web-based solution to improve the accuracy, efficiency, and reliability of petroleum shore tank inspection documentation and reporting. The system will reduce manual errors, speed up the preparation of inspection reports and enable faster petroleum quantity reconciliation across depots. Additionally, centralized and secure digital storage will enhance data accessibility, traceability, regulatory compliance, and decision-making. Overall, the study supports PBPA's digital transformation and offers a scalable approach for modernizing petroleum inspection systems in Tanzania.

CHAPTER TWO

2. LITERATURE REVIEW

2.1 Introduction

This chapter presents a comprehensive review of literature related to the design and development of a smart web-based documentation and reporting system for petroleum shore tank inspections. The review establishes the theoretical, conceptual, and empirical foundations of the study and critically examines existing scholarly works relevant to petroleum inspection systems, management information systems (MIS), web-based reporting platforms, database technologies, and information security. The chapter concludes with an appraisal of the reviewed literature, identifying research gaps that justify the present study.

2.2 Theoretical framework of the study

A theoretical framework provides the underlying theories that explain how and why a proposed system improves organizational processes. This study is anchored on four main theories: Management Information Systems (MIS) Theory, Socio-Technical Systems Theory, Information Quality Theory, and Technology Acceptance Model (TAM).

2.2.1 Management Information Systems (MIS) Theory.

Management Information Systems theory focuses on the use of information systems to support managerial decision-making, operational control, and organizational efficiency. According to (Laudon, 2016), MIS integrates people, processes, data, and technology to produce timely and accurate information for decision-making. In petroleum operations, MIS plays a critical role in managing large volumes of operational data such as inspection measurements, inventory levels, and compliance records.

In the context of PBPA shore tank inspections, a smart web-based reporting system functions as an MIS by collecting inspection data, processing it through automated calculations, and generating standardized reports. Recent studies indicate that digital MIS platforms significantly enhance data consistency, transparency, and accountability in resource intensive industries such as oil and gas (Deo, 2023) Therefore, MIS theory supports the justification that a web-based system can improve inspection reporting accuracy and managerial oversight at PBPA.

2.2.2 Social-Technical Systems Theory.

Social-Technical Systems (STS) theory emphasizes the interdependence between social components (users, organizational structures, workflows) and technical components (software, databases, hardware) of an information system. The theory argues that optimal system performance is achieved when both social and technical subsystems are jointly optimized (Baxter, 2011)

Applying STS theory to this study highlights the importance of designing a system that aligns with PBPA inspector's workflows, competencies, and regulatory responsibilities. A smart documentation and reporting system must not only provide technical capabilities such as automated calculations and secure storage but also support inspector's usability needs, collaboration, and accountability. Recent research in inspection systems confirms that failure to consider social-technical factors often leads to system resistance and underutilization (Sarker, (2021).)

2.2.3 Information Quality Theory

Information Quality (IQ) theory focuses on the characteristics of data that make it fit for use, including accuracy, completeness, timeliness, consistency, and reliability. Strong information quality is essential in petroleum inspection activities, where small measurement errors can lead to significant financial and operational consequences (Eppler, 2023)

Manual documentation practices often compromise information quality due to transcription errors, inconsistent formats, and delaying reporting. A Web-based systems improve information quality by enforcing data validation rules, standardized data entry forms, and automated computations. Recent studies show that digital inspection systems significantly enhance data accuracy and auditability in regulated industries (Sotnik, 2023). Thus, IQ theory underpins the need for structured digital documentation for PBPA inspections.

2.2.4 Technology Acceptance Model (TAM).

The Technology Acceptance Model explains user adoption of information systems based on perceived usefulness and perceived ease of use (Davis, 1989). For PBPA inspectors to effectively adopt a smart reporting system, the system must demonstrably improve inspection efficiency and be easy to use in operational environments.

Recent extensions of TAM in government and industrial systems reveal that system reliability, security, and compatibility with existing workflows significantly influence user

acceptance (Al-Emran, 2023). Therefore, TAM provides a theoretical basis for designing a user-friendly and reliable system that inspectors are willing to adopt.

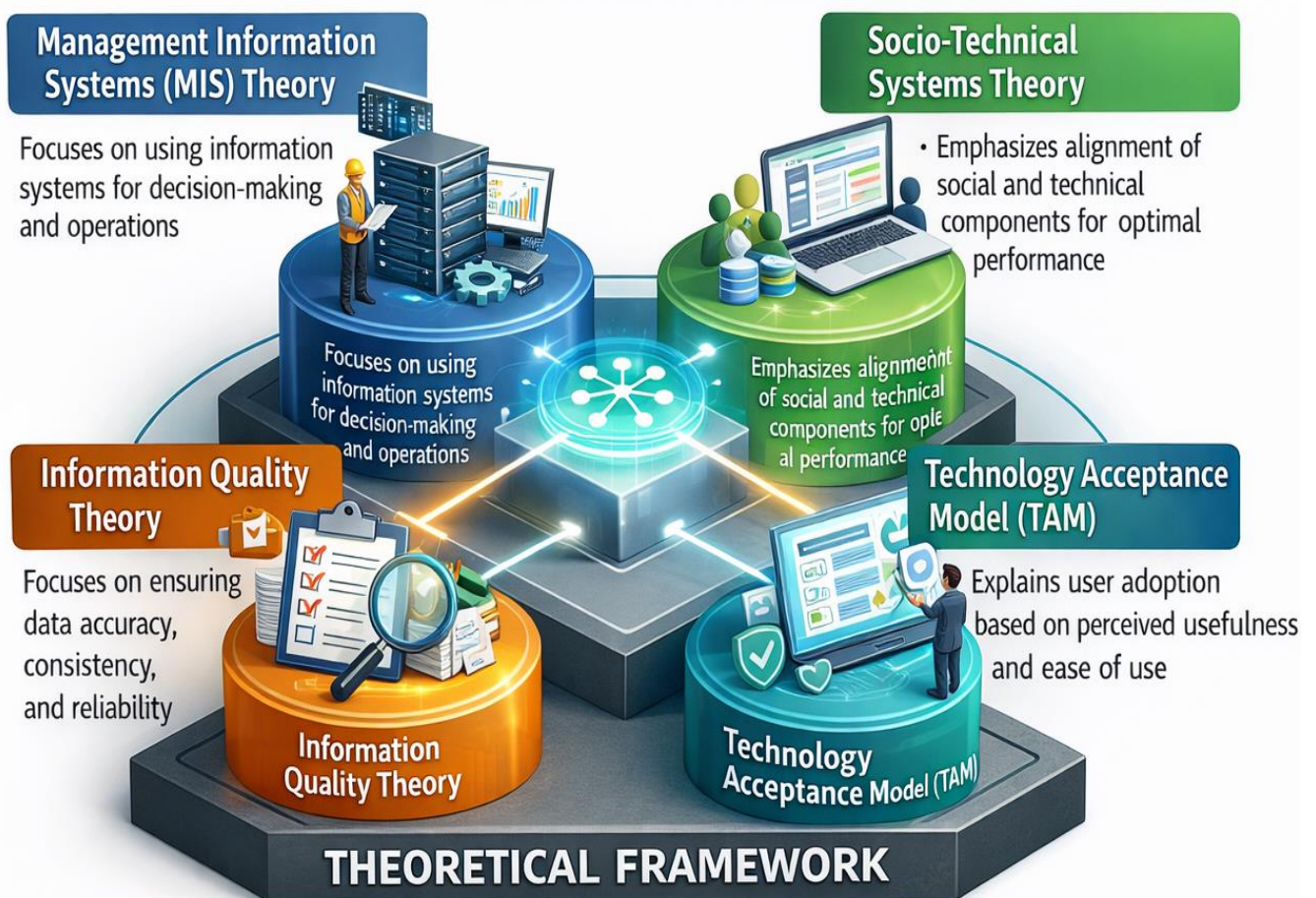


Figure 1: Theoretical framework

Source: AI generated

2.3 Conceptual framework

The conceptual framework illustrates the relationship between system inputs, processes, and outputs in the proposed smart web-based documentation and reporting system. It explains how digital inspection data is transformed into reliable reports that support PBPA's operational and managerial objectives.

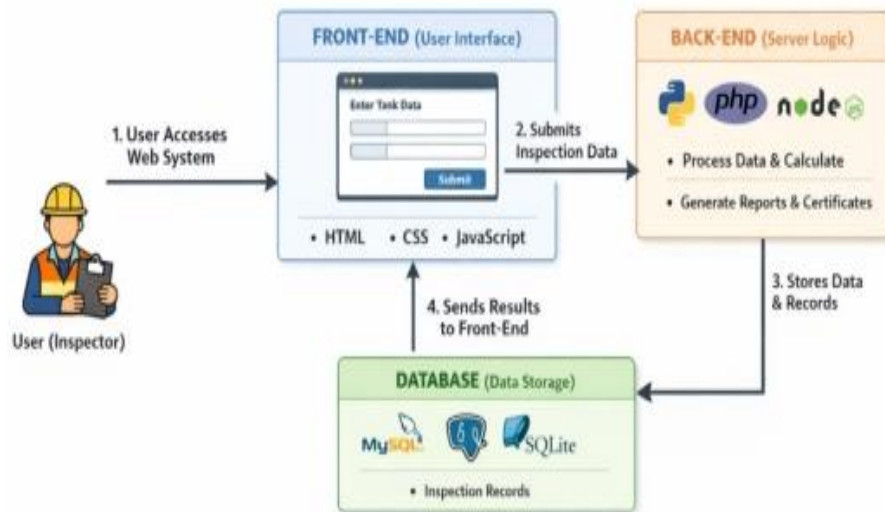


Figure 2: Conceptual structure of the system

2.4 Empirical review.

Empirical review presents findings from previous studies related to the development and implementation of web-based documentation and reporting systems, database-driven applications, petroleum quantity automation, and information security in digital platforms. The purpose of this section is to examine how similar systems have been applied in industrial and public sector environments, identify best practices, and highlight gaps that justify the need for the proposed smart web-based documentation and reporting system for petroleum shore tank inspections at PBPA. The reviewed studies provide evidence on the effectiveness of digital inspection systems in improving data accuracy, operational efficiency, transparency, and decision-making processes.

2.4.1 Web-Based Inspection and Reporting Systems.

Empirical studies show that web-based inspection systems significantly improve efficiency and data accuracy in industrial inspections. (Deo, 2023) found that e-reporting systems

reduced report preparation time by over 40% in regulated environments. Similarly, (Zhang, 2022) reported improved traceability and audit readiness in digital inspection platforms used in energy infrastructure.

2.4.2 Database-Driven Documentation Systems.

MySQL and relational database systems are widely adopted for managing large-scale operational data. (Appigatla, 2018) and (Sotnik, 2023) demonstrated that relational databases support high-performance querying, data integrity, and scalability in web-based applications. In petroleum operations, centralized databases enhance consistency across geographically distributed sites such as ports and depots.

2.4.3 Automation of Petroleum Quantity Calculations.

Studies in petroleum logistics indicate that automated volume and mass calculations reduce reconciliation discrepancies and losses (Al-Fatlawi,2021) Digital calculation engines embedded in inspection systems eliminate manual formula errors and ensure compliance with international measurement standards.

2.4.4 Information Security in Web-Based Systems.

Inspection data is sensitive and requires strict security controls. Research by (Behl, 2022.) emphasizes the importance of role-based access control, encryption, and audit trails in government information systems. Recent studies confirm that secure web architectures significantly reduce data breaches and unauthorized access (Khan, 2023.)

2.4.5 Adoption of Digital Systems in Public Sector Organizations

Empirical evidence shows increasing adoption of digital reporting systems in public agencies to enhance transparency and efficiency. (Al-Emran, 2023) highlight that system usability, trust, and perceived benefits strongly influence adoption success. These findings are directly applicable to PBPA's operational environment

CHAPTER THREE

3. METHODOLOGY

3.1 Introduction

This chapter describes the methodology adopted to achieve the objectives of this study, which focuses on designing and developing a smart web-based documentation and reporting system to support petroleum shore tank inspection activities conducted by the Petroleum Bulk Procurement Agency (PBPA). The study employed the Agile software development methodology due to its iterative, flexible, and user-centred nature, which allows continuous system improvement based on feedback from stakeholders. Agile methodology is widely recommended for information system development projects where requirements evolve and user involvement is essential (Pressman, 2020) (Sommerville, 2016)

The chapter explains the procedures used to analyse existing inspection processes, design and develop the proposed system, and test the automated reporting module. Each section corresponds to one of the project objectives to ensure methodological clarity and alignment with the study goals.

3.2 Analysis of existing manual processes for Petroleum shore tank inspection operations at PBPA.

The first objective of this study will be the analysis of the existing manual processes used by PBPA field engineers during petroleum shore tank inspections. A qualitative system analysis approach will be employed to understand the current workflow, challenges, and user requirements. Data will be collected using the following methods;

3.2.1 Interviews

Semi-structured interviews will be conducted with PBPA inspectors to obtain information regarding inspection procedures, documentation practices, and challenges experienced during reporting activities. Interviews are widely used in system analysis to gather user requirements and operational insights (Kothari, 2019)

3.2.2 Field Observation

Direct observation of inspection activities will be performed to understand how measurements will be recorded, calculated, and reported during petroleum shore tank inspections.

Observation will provide reliable information about actual operational practices and will help identify inefficiencies associated with the existing manual system.

3.2.3 Document Review

Existing inspection templates and reporting documents will be analysed to identify the data elements, reporting formats, and calculation procedures that will be used in the current system.

3.2.4 Process Mapping

Workflow diagrams will be developed to visualize the current inspection reporting process and to identify inefficiencies such as data duplication, delays, and calculation errors. Process mapping will support requirement identification and system improvement (Dennis, 2018)

The findings from this phase will be used to create the product backlog (Development Requirement List), ensuring that the proposed system features will address real operational needs.

3.3 System design and development

The second objective will be to design and develop a smart web-based system that will enable accurate digital capture and secure management of petroleum shore tank inspection data. The Agile methodology will be applied because it will support incremental delivery, continuous testing, and integration of user feedback (Sommerville, 2016)

The development process will follow the Agile phases described below;

3.3.1 Requirements Backlog Development

System requirements identified during the analysis phase will be converted into user stories and prioritized in a project backlog. The backlog will include modules such as:

- i. User authentication and access control
- ii. Inspection data entry forms
- iii. Automated petroleum quantity calculations
- iv. Report generation
- v. Database management
- vi. Security and audit logging

3.3.2 Sprint Planning and Iterative Development

Development tasks will be divided into iterative cycles (sprints). Each sprint will deliver a functional system component, allowing early testing and feedback. Agile iteration will improve software quality and user satisfaction by enabling continuous refinement. (Pressman, 2020)

3.3.3 System Architecture Design

The proposed system will follow a three-tier architecture consisting of:

- i. User Interface (Presentation Layer)
- ii. Server Logic (Application Layer)
- iii. Database (Data Layer)

This architecture will improve scalability, maintainability, and security in web-based systems (Sommerville, 2016)

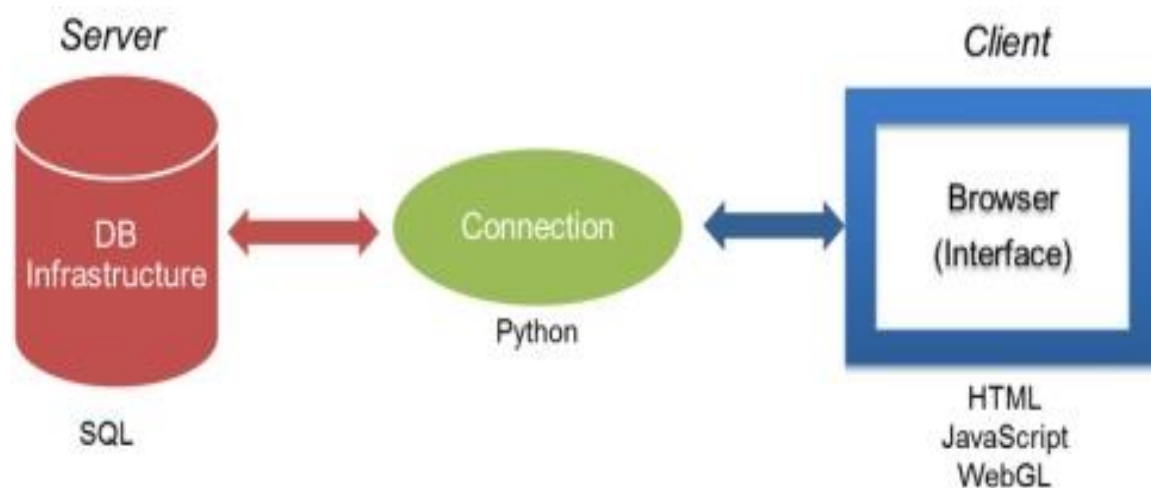


Figure 3 System Architecture Design

Source: Adapted from (Sommerville,2016)

The architecture will show how inspectors will interact with the system through web browsers, while the application server will process data and will communicate with the centralized database.

3.3.4 System Design Tools and Technologies

The system will be developed using modern web technologies:

- i. Frontend: HTML, CSS, JavaScript
- ii. Backend: PHP and Python
- iii. Database: MySQL
- iv. Development Tools: Visual Studio Code (VS code).

Relational databases such as MySQL will support efficient storage and retrieval of structured operational data (Appigatla, 2018)

3.3.5 Automated Calculation Module.

The system will integrate automated petroleum quantity calculations to eliminate manual computation errors. Automated systems will improve accuracy and reconciliation reliability in petroleum measurement processes (Al-Fatlawi, 2021)

3.4 System testing and automated reporting module

The third objective will be to test an automated reporting module based on inspection measurements. Agile testing methodology will be applied to ensure continuous quality improvement during development.

3.4.1 Unit Testing

Individual modules such as login, data entry, and calculations will be tested independently to verify correctness.

3.4.2 Integration Testing

Modules will be combined and tested to ensure proper interaction between system components.

3.4.3 User Acceptance Testing (UAT)

Testing will be conducted using sample petroleum inspection datasets to validate system performance. User testing will improve system acceptance and effectiveness. (Al-Emran, 2023)

3.4.4 Validation of Automated Reports

Generated reports will be compared with manually prepared reports to verify its accuracy and reliability.

3.4.5 Agile Review and Feedback

Feedback from testing sessions will be incorporated into subsequent development iterations, ensuring continuous system improvement.

The Agile cycle illustrates iterative development, testing, and feedback integration throughout the project.

3.4.6 Expected Results

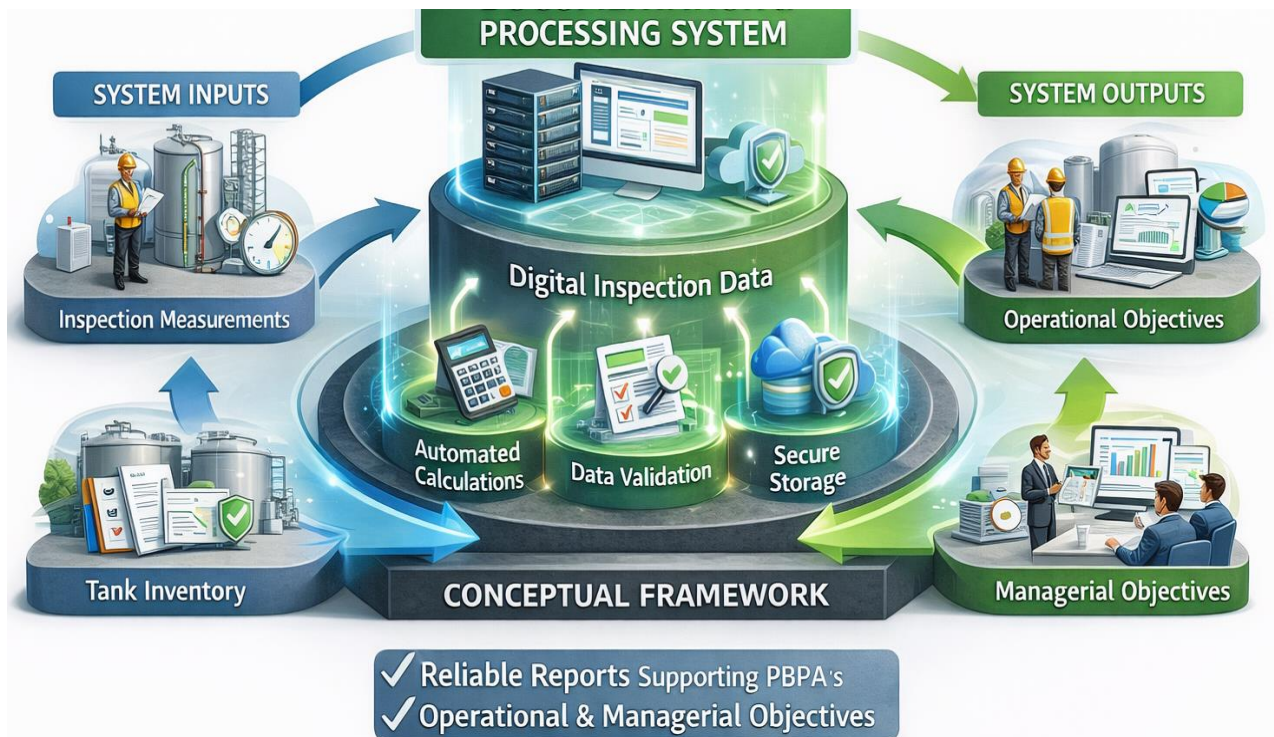


Figure 4:Expected Results

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APPENDICES

NOTE: Here you are expected to provide any supplementary information related to what you have stated within your report. Do not just attach anything, make sure it is clearly useful and stated within your report (i.e. cross-referenced)